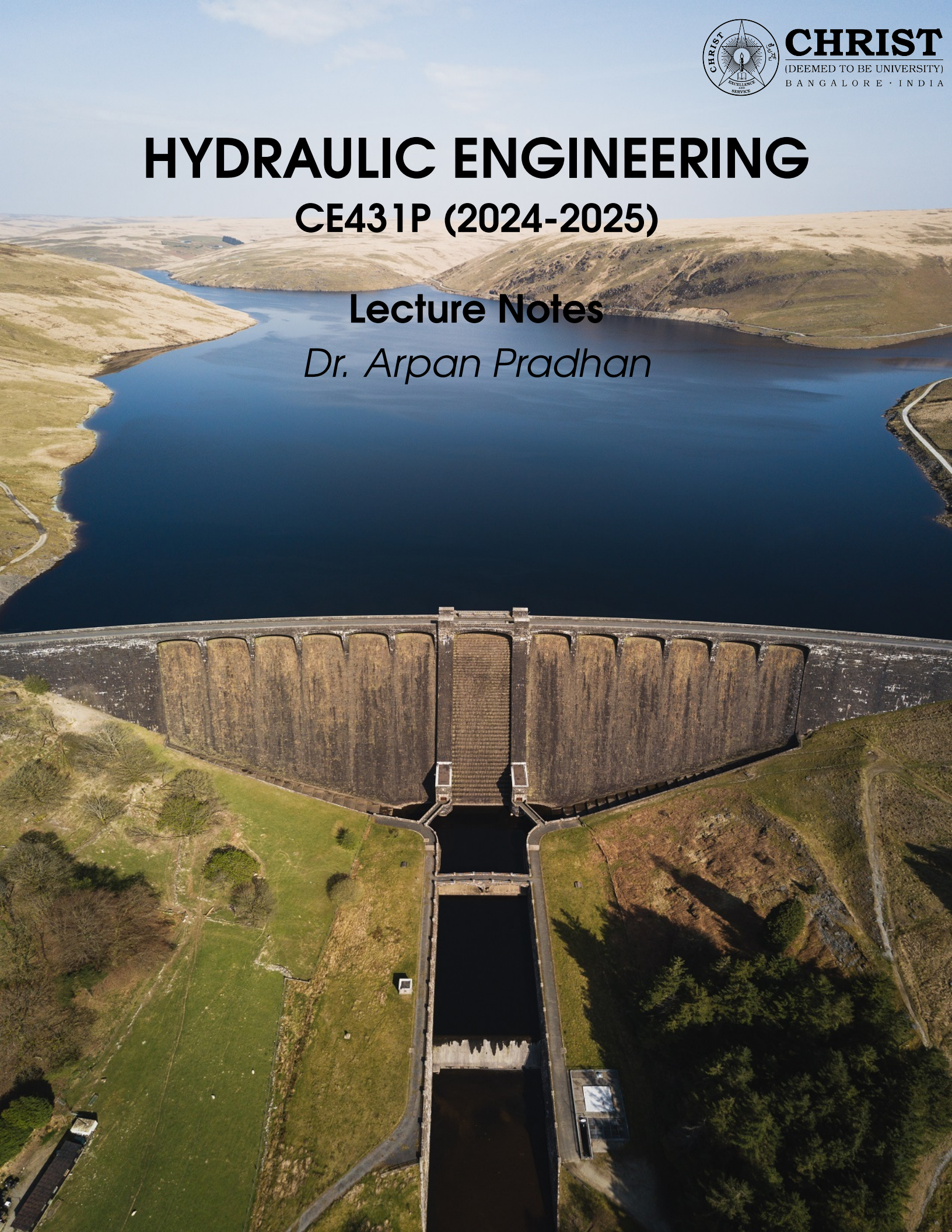
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HYDRAULIC ENGINEERING

CE431P (2024-2025)

Lecture Notes

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Contents

1 Computational Fluid Dynamics 5

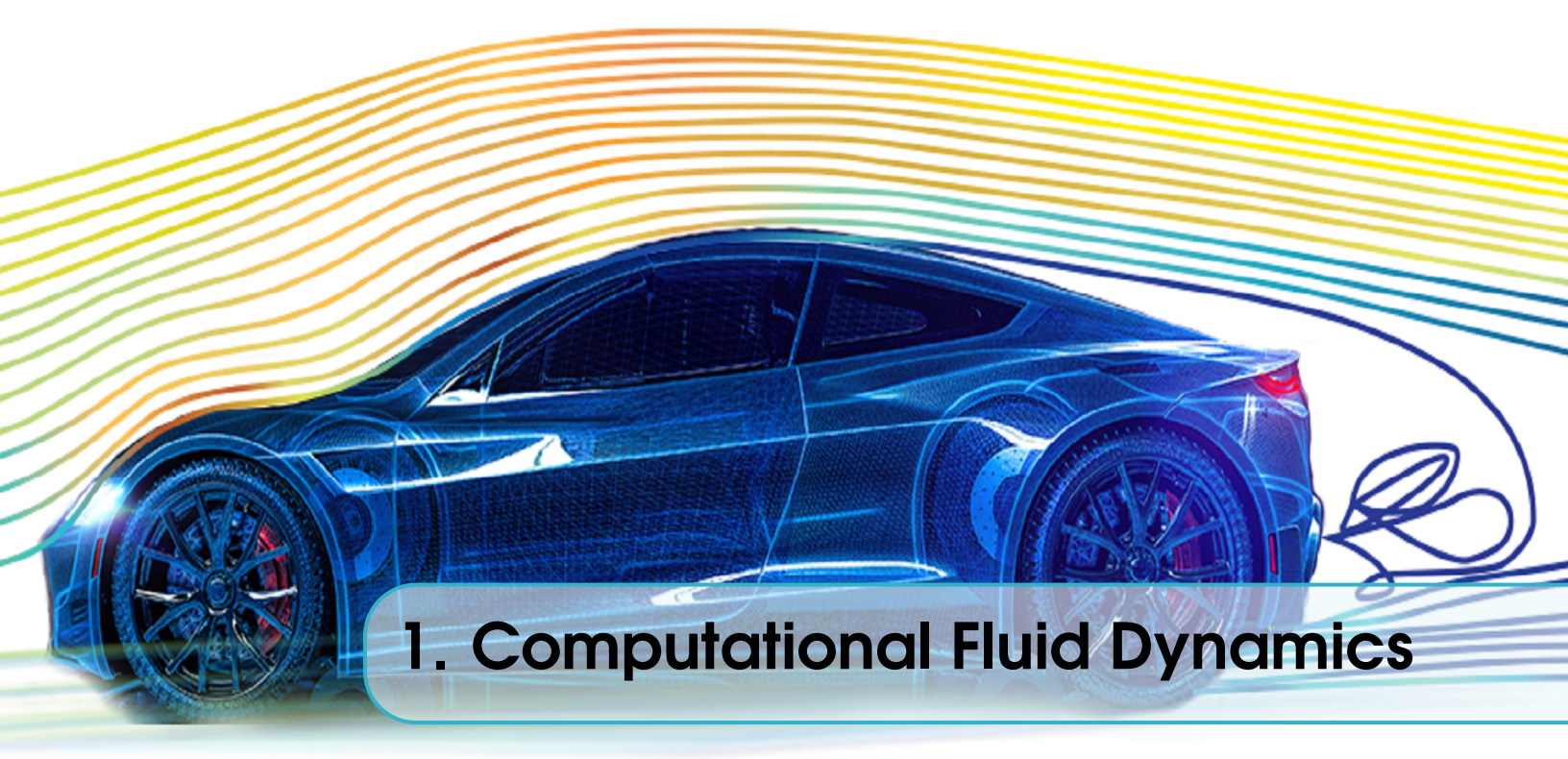
1.1 Introduction to Computational Fluid Dynamics 5
1.1.1 Definition 5
1.1.2 Scope of CFD 5
1.1.3 Importance of CFD in Fluid Flow Analysis 6
1.1.4 Comparison with Experimental and Analytical Methods 6
1.1.5 Fundamental Concepts in CFD Modeling 7

1.2 Simplified forms of the Navier-Stokes equations 9
1.2.1 Euler Equations 9
1.2.2 Reynolds-Averaged Navier-Stokes (RANS) Equations 9
1.2.3 Conclusion 10

1.3 Numerical Methods for CFD 11
1.3.1 Finite Difference Method (FDM) 11
1.3.2 Finite Volume Method (FVM) 11
1.3.3 Finite Element Method (FEM) 11

1.4 Turbulence Modeling in Fluid Dynamics 13
1.4.1 Nature of Turbulence: Unsteady, Chaotic Motion 13
1.4.2 Importance of Turbulence Modeling 13
1.4.3 Reynolds-Averaged Navier-Stokes (RANS) Models 14
1.4.4 Large Eddy Simulation (LES) 14
1.4.5 Detached Eddy Simulation (DES) 14
1.4.6 Comparison of Different Turbulence Models 15
1.4.7 Conclusion 15

1.5 Comparison of Grids in CFD Simulations 16
1.5.1 Structured Grid 16
1.5.2 Unstructured Grid 16
1.5.3 Hybrid Grid 17
1.5.4 Conclusion 17
1.6 Experimental Validation and Calibration in CFD 18
1.6.1 Importance of Experimental Data in CFD 18
1.6.2 Need for Validation Using Experimental Results 18
1.6.3 Sources of Experimental Data 18
1.7 Basics of Modeling Using HEC-RAS 20
1.7.1 Introduction to HEC-RAS 20
1.7.2 Key Features of HEC-RAS 20
1.7.3 Steps in HEC-RAS Modeling 20
1.7.4 Model Calibration and Validation 21
1.7.5 Applications of HEC-RAS 21
1.7.6 Future Trends in HEC-RAS Modeling 22
1.8 Uniform and Non-Uniform Flow in Open Channel Simulations 23
1.8.1 Uniform Flow: 23
1.8.2 Non-Uniform Flow: 23
1.8.3 Using HEC-RAS for Uniform and Non-Uniform Flow Simulation 23
1.8.4 Example: Simulating Uniform and Non-Uniform Flow in HEC-RAS 23



1. Computational Fluid Dynamics

"Nothing is softer or more flexible than water, yet nothing can resist it."

- Lao Tzu

1.1 Introduction to Computational Fluid Dynamics

1.1.1 Definition

Computational Fluid Dynamics (CFD) is a field of fluid mechanics that utilizes numerical methods and algorithms to analyze fluid flow problems. It involves solving the fundamental governing equations of fluid motion, including the Navier-Stokes equations, continuity equations, and energy equations, using computational techniques. CFD is widely used across various industries, including aerospace, automotive, civil engineering, and biomedical applications. Computational Fluid Dynamics (CFD) is a specialized discipline within fluid mechanics that employs numerical methods and algorithms to solve and analyze problems involving fluid flows. It enables engineers and scientists to simulate complex fluid dynamics phenomena within a computational framework, reducing reliance on costly and time consuming experimentation.

CFD integrates fundamental physics, mathematics, and computational sciences to provide detailed insights into fluid behavior. The governing equations of CFD, primarily the **Navier-Stokes equations**, describe the conservation of mass, momentum, and energy in a fluid flow system. These equations are solved using advanced numerical techniques and computational power.

1.1.2 Scope of CFD

CFD has extensive applications across multiple engineering and scientific fields, including:

- **Aerospace Engineering**:
  Aerodynamic analysis of aircraft, jet propulsion modeling, and turbulence studies.
- **Automotive Industry**:
  Vehicle aerodynamics optimization, internal combustion engine simulations, and cooling system design.
- **Civil and Environmental Engineering**:
  River hydraulics modeling, flood forecasting, and pollutant dispersion analysis.
- **Biomedical Engineering**:
  Blood flow modeling in arteries, prosthetic device optimization, and respiratory airflow studies.
- **Marine Engineering**:
  Ship hull design, offshore structure analysis, and underwater vehicle simulations.
- **Energy Sector**:
  Wind turbine performance analysis, heat exchanger optimization, and nuclear reactor thermal management.

1.1.3 Importance of CFD in Fluid Flow Analysis

CFD provides detailed insight into fluid behavior without the need for expensive and time-consuming physical experiments. It helps engineers optimize designs, evaluate system performance, and predict operational challenges.

- **Optimization of Designs**:
  Enhances the performance of engineering systems such as aircraft wings, turbines, and heat exchangers.
- **Performance Evaluation**:
  Assesses the behavior of hydraulic structures, industrial cooling systems, and HVAC systems.
- **Safety Analysis**:
  Helps predict fire and smoke movement in buildings, crucial for emergency planning.
- **Environmental Impact Studies**:
  Models pollutant transport in air and water bodies to assess ecological impacts.
- **Medical Advancements**:
  Assists in the design of medical implants, drug delivery systems, and fluid-based diagnostics.

1.1.4 Comparison with Experimental and Analytical Methods

CFD serves as a complementary tool to traditional experimental and analytical methods. Below is a comparative analysis:

Table 1.1: Comparison of CFD, Experimental, and Analytical Methods

Aspect	**CFD**	**Experimental Methods**	**Analytical Methods**
Cost	Lower after setup	High due to laboratory expenses	Moderate
Time	Shorter simulation time	Time-intensive setup and data collection	Case-dependent
Data Coverage	Provides full-field data	Limited to measurement locations	Restricted by assumptions
Flexibility	High, adaptable to different conditions	Difficult due to physical constraints	Limited to solvable equations
Accuracy	Dependent on models and discretization	High but subject to instrumentation errors	Limited by theoretical assumptions

1.1.5 Fundamental Concepts in CFD Modeling

- ➡ Governing Equations of Fluid Flow
- ➡ Numerical Methods for CFD
- ➡ Turbulence Modeling in CFD
- ➡ Mesh Generation in CFD

Governing Equations of Fluid Flow

CFD simulations rely on fundamental fluid mechanics equations:

- ➡ **Continuity Equation**:
  Represents mass conservation.
- ➡ **Navier-Stokes Equations**:
  Governs fluid motion considering inertia, pressure, viscosity, and external forces.
- ➡ **Energy Equation**:
  Accounts for heat transfer and temperature changes within the fluid.

Numerical Methods for CFD

Numerical discretization is essential for solving CFD problems:

- ➡ **Finite Difference Method (FDM)**:
  Uses numerical differentiation to approximate derivatives.
- ➡ **Finite Volume Method (FVM)**:
  Solves integral forms of governing equations.
- ➡ **Finite Element Method (FEM)**:
  Employs variational principles for complex geometries.

Turbulence Modeling in CFD

Turbulence plays a critical role in fluid flow simulations, necessitating specialized models:

- **Direct Numerical Simulation (DNS)**:
  Resolves all turbulence scales (computationally intensive).
- **Large Eddy Simulation (LES)**:
  Captures large turbulent structures while modeling smaller ones.
- **Reynolds-Averaged Navier-Stokes (RANS)**:
  Uses statistical averaging for engineering applications.

Mesh Generation in CFD

Mesh quality significantly impacts simulation accuracy:

- **Structured Mesh**:
  Regular grid with high precision but less flexibility.
- **Unstructured Mesh**:
  Adaptable to complex geometries but computationally expensive.
- **Hybrid Mesh**:
  Balances accuracy and efficiency for intricate domains.

1.2 Simplified forms of the Navier-Stokes equations

The Navier-Stokes (NS) equations describe the motion of fluid flow, governing conservation of mass, momentum, and energy. However, their full form is highly complex and often requires simplifications for practical Computational Fluid Dynamics (CFD) simulations. Two common simplified forms are the Euler equations and the Reynolds-Averaged Navier-Stokes (RANS) equations.

1.2.1 Euler Equations

Assumptions

- **Inviscid flow**: Neglects viscosity ($\mu = 0$), meaning no shear stresses.
- **Irrotational flow**: Assumes no vorticity (though not always required).
- **No turbulence modeling**: Does not account for small-scale fluctuations.

Derivation

Starting from the Navier-Stokes equations, the momentum equation is:

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\frac{1}{\rho}\nabla p + \nu \nabla^2 u + g \tag{1.1}$$

Setting viscosity $\nu = 0$ (inviscid assumption) removes the viscous term ($\nu \nabla^2 u$), leading to:

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\frac{1}{\rho}\nabla p + g \tag{1.2}$$

Additionally, the continuity equation remains:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0 \tag{1.3}$$

Application in CFD

- Used for high-speed aerodynamics and compressible flows where viscous effects are negligible.
- Often applied in external flows, such as supersonic/hypersonic aircraft design.
- Less accurate for boundary layers and turbulent regions due to the neglect of viscosity.

1.2.2 Reynolds-Averaged Navier-Stokes (RANS) Equations

Assumptions

- **Turbulent flow decomposition**: Velocity is decomposed into mean and fluctuating components:

$$u = \overline{u} + u' \tag{1.4}$$

where $\overline{u}$ is the time-averaged velocity and u' is the fluctuating part.

- ➡ **Reynolds averaging**: Time-averaging removes small turbulent fluctuations.
- ➡ **Closure models required**: Since turbulence introduces unknowns, additional models (like k-ε or k-ω) are needed.

Derivation

Reynolds averaging the Navier-Stokes momentum equation gives:

$$\frac{\partial \overline{u}}{\partial t} + (\overline{u} \cdot \nabla)\overline{u} = -\frac{1}{\rho}\nabla \overline{p} + \nu \nabla^2 \overline{u} - \nabla \cdot \overline{u'u'} \tag{1.5}$$

where the term $-\nabla \cdot \overline{u'u'}$ represents the Reynolds stress tensor, which must be modeled using turbulence models.

Application in CFD

- ➡ Widely used in engineering applications like aircraft design, HVAC systems, and environmental flows.
- ➡ Provides an efficient way to model turbulent flows without resolving every small fluctuation (as in Direct Numerical Simulation, DNS).
- ➡ Used in steady-state and unsteady CFD simulations, but requires careful selection of turbulence models (e.g., k-ε, k-ω).

1.2.3 Conclusion

- ➡ Euler equations are useful for high-speed, inviscid flows but cannot capture turbulence or boundary layer effects.
- ➡ RANS equations simplify turbulent flows using averaging, making them ideal for industrial and engineering applications.
- ➡ The choice depends on the problem: Euler for inviscid high-speed flow, RANS for practical turbulent flow modeling.

1.3 Numerical Methods for CFD

Numerical discretization is fundamental in Computational Fluid Dynamics (CFD) as it enables the transformation of governing differential equations into algebraic equations that can be solved using computational methods. Several numerical techniques exist, each with distinct advantages and applications. The most commonly used methods in CFD include:

1.3.1 Finite Difference Method (FDM)

The Finite Difference Method (FDM) is based on approximating the derivatives of the governing partial differential equations (PDEs) using numerical differentiation. The domain is discretized into a structured grid, and derivatives are estimated using difference equations such as forward, backward, or central differences.

Advantages:

1. Simple to implement, especially for structured grids.
2. Computationally efficient for problems with regular geometries.

Disadvantages:

1. Less flexible for complex geometries. Accuracy depends on the choice of grid spacing and numerical scheme (explicit vs. implicit).
2. FDM is widely used in problems involving Cartesian grids and for solving problems such as boundary layer flows and turbulence modeling.

1.3.2 Finite Volume Method (FVM)

The Finite Volume Method (FVM) discretizes the governing equations in their integral form, ensuring conservation of mass, momentum, and energy over control volumes. The domain is divided into small control volumes (cells), and fluxes across the cell faces are evaluated using interpolation and numerical integration techniques.

Advantages:

1. Strict conservation properties, making it suitable for fluid flow problems.
2. Flexibility in handling unstructured grids and complex geometries.

Disadvantages:

1. More computationally intensive compared to FDM.
2. Requires careful selection of interpolation and flux reconstruction schemes.
3. FVM is extensively used in commercial CFD solvers such as ANSYS Fluent and OpenFOAM due to its robustness in solving Navier-Stokes equations.

1.3.3 Finite Element Method (FEM)

The Finite Element Method (FEM) employs variational principles and weighted residual methods to approximate solutions. It is particularly useful for solving problems involving complex geometries and irregular boundaries. The computational domain is divided into

elements, and shape functions (basis functions) are used to approximate the solution within each element.

Advantages:

1. Excellent for handling complex geometries.
2. Can accommodate higher-order accuracy through polynomial basis functions.

Disadvantages:

1. Computationally expensive due to matrix assembly and inversion.
2. More complex implementation compared to FDM and FVM.

FEM is commonly used in structural mechanics and multiphysics simulations but is also applied in CFD, particularly for problems involving porous media and fluid-structure interactions.

Each of these methods has its strengths and limitations, and the choice of method depends on the specific nature of the CFD problem, the desired accuracy, and computational efficiency.

1.4 Turbulence Modeling in Fluid Dynamics

Turbulence is a complex, chaotic state of fluid motion characterized by vortices, eddies, and fluctuating velocity fields. It occurs in most real-world flows, including air around an aircraft, water in rivers, and wind in urban environments.

Solving the **Navier-Stokes equations** directly for turbulent flows requires resolving all scales of motion, which is computationally infeasible for most practical problems. This is why **turbulence modeling** is necessary—to approximate turbulent effects while keeping computations manageable.

1.4.1 Nature of Turbulence: Unsteady, Chaotic Motion

Turbulence is a complex phenomenon in fluid dynamics characterized by irregular, three-dimensional vortices and rapid variations in velocity and pressure. It occurs when the inertial forces in a fluid flow dominate over viscous forces, leading to an unpredictable and chaotic motion of fluid particles.

Turbulence commonly occurs in high Reynolds number flows, where:

- There is a significant difference in velocity between fluid layers.
- Flow interactions create swirling eddies of different scales.
- Energy cascades from large vortices to smaller ones until it dissipates as heat.

Examples of turbulence in real-world applications include:

- **Airflow over an aircraft wing:**
  The boundary layer transitions from laminar to turbulent, affecting lift and drag forces.
- **River flows:**
  Rapid and turbulent currents occur around obstacles like boulders, bridge piers, and bends.
- **Combustion in engines:**
  Turbulent mixing of fuel and air enhances combustion efficiency.
- **Atmospheric turbulence:**
  Wind patterns create turbulence in weather systems, affecting cloud formation and dispersion of pollutants.
- **Turbulent mixing in oceans:**
  Plays a crucial role in nutrient distribution and energy transfer.

1.4.2 Importance of Turbulence Modeling

Turbulence modeling is essential for accurate simulations in engineering and environmental studies due to the following reasons:

- **Predicting Flow Behavior:**
  Understanding how turbulence affects velocity profiles, pressure distribution, and shear stresses is critical for design and performance evaluation.
- **Enhancing Simulation Accuracy:**
  Direct Numerical Simulation (DNS) of turbulence is computationally expensive, making turbulence models necessary for practical applications.

- ➡ **Designing Efficient Systems:**
  Engineering applications, such as aircraft aerodynamics, hydraulic structures, and HVAC systems, require turbulence modeling to optimize performance.
- ➡ **Safety and Reliability:**
  Accurate turbulence prediction helps mitigate risks associated with extreme weather conditions, fluid-structure interactions, and industrial processes.

Turbulence modeling remains a fundamental aspect of computational fluid dynamics (CFD), enabling engineers and scientists to analyze and optimize fluid flow behavior in complex environments.

1.4.3 Reynolds-Averaged Navier-Stokes (RANS) Models

k-ε Model

- ➡ Solves two transport equations (turbulent kinetic energy k and dissipation rate ε).
- ➡ **Pros:** Robust, widely used for industrial flows.
- ➡ **Cons:** Struggles with flow separation and near-wall accuracy.
- ➡ **Application:** HVAC, pipelines, hydrodynamics, civil structures.

k-ω Model

- ➡ Uses ω (specific dissipation rate) instead of ε.
- ➡ **Pros:** Better for boundary layers and low-Re flows.
- ➡ **Cons:** More sensitive to freestream turbulence conditions.
- ➡ **Application:** Aerospace, marine hydrodynamics, wind tunnels.

Shear Stress Transport (SST) Model

- ➡ Hybrid of k-ε and k-ω uses k-ω near walls and k-ε in free-stream regions.
- ➡ **Pros:** Improved accuracy for separation and adverse pressure gradients.
- ➡ **Cons:** More computationally expensive than standard RANS.
- ➡ **Application:** Airfoil aerodynamics, ship hulls, turbomachinery.

1.4.4 Large Eddy Simulation (LES)

- ➡ Resolves large turbulence structures while modeling small-scale effects using a sub-grid scale (SGS) model.
- ➡ **Pros:** Captures unsteady turbulence more accurately than RANS.
- ➡ **Cons:** Requires fine meshes and high computational power.
- ➡ **Application:** Jet engines, combustion modeling, meteorology, aerospace.

1.4.5 Detached Eddy Simulation (DES)

- ➡ Combines LES for large turbulent structures and RANS near walls.
- ➡ **Pros:** Reduces computational cost compared to full LES.
- ➡ **Cons:** Less accurate than pure LES for some flows.
- ➡ **Application:** Car aerodynamics, high-Re flows, bluff body flows.

1.4.6 Comparison of Different Turbulence Models

Table 1.2: Comparison of Turbulence Models

Model	Description	Applicability	Computational Cost
Direct Numerical Simulation (DNS)	Solves all turbulence scales directly.	Only for fundamental research ($Re < 1000$).	Extremely high (prohibitive for practical use).
Large Eddy Simulation (LES)	Resolves large turbulence scales, models small ones.	High-Re flows, external aerodynamics, combustion.	High (but lower than DNS).
Reynolds-Averaged Navier-Stokes (RANS)	Models all turbulence effects using statistical averaging.	Engineering applications (pipes, aircraft, HVAC).	Low (widely used in industry).
Detached Eddy Simulation (DES)	Hybrid LES-RANS approach; LES in unsteady regions, RANS elsewhere.	Aeroacoustics, external flows, car aerodynamics.	Moderate (less expensive than LES).

1.4.7 Conclusion

- ➡ For industrial flows (pipes, buildings, cars) $\rightarrow$ Use RANS models (k-ε, SST).
- ➡ For high-fidelity aerodynamics or turbulence studies $\rightarrow$ Use LES or DES.
- ➡ For fundamental research in turbulence $\rightarrow$ Use DNS (only feasible for low Reynolds numbers).

The choice of turbulence model depends on the trade-off between accuracy and computational efficiency.

1.5 Comparison of Grids in CFD Simulations

In Computational Fluid Dynamics (CFD), grid generation is crucial for accurate simulations. While OpenFOAM primarily uses structured, unstructured, and hybrid grids for CFD applications, HEC-RAS also employs different grid types for hydraulic modeling, particularly in its 2D and unsteady flow simulations.

1.5.1 Structured Grid

Description:

Consists of regular, evenly spaced grids (typically quadrilateral or hexahedral cells).

Advantages:

- ➡ More accurate and computationally efficient due to alignment with flow direction.
- ➡ Requires less memory and processing power.
- ➡ Easier to apply numerical schemes with reduced numerical diffusion.

Disadvantages:

- ➡ Difficult to use for complex geometries like river meanders or floodplains.
- ➡ Requires significant pre-processing to fit irregular boundaries.

Use in HEC-RAS:

- ➡ Typically not used for 1D simulations.
- ➡ 2D flow areas in HEC-RAS can have structured grids, though it is less common.

1.5.2 Unstructured Grid

Description:

Composed of irregularly shaped elements (triangular or tetrahedral in 2D/3D).

Advantages:

- ➡ Suitable for complex geometries, such as rivers with varying cross-sections and floodplain expansions.
- ➡ Can adapt to changes in resolution, refining the grid where necessary.

Disadvantages:

- ➡ Higher computational cost due to irregular connectivity of nodes.
- ➡ Can introduce numerical diffusion if not properly refined.

Use in HEC-RAS:

- ➡ Used extensively in 2D modeling, allowing flexible meshing of rivers, floodplains, and structures.
- ➡ Adaptive refinement helps in areas with rapid changes in flow characteristics.

1.5.3 Hybrid Grid

Description:

A combination of structured and unstructured grids, using structured grids in simple regions and unstructured grids in complex areas.

Advantages:

- ➡ Balances accuracy and computational efficiency.
- ➡ Allows structured grids in uniform regions and unstructured grids where high resolution is needed.

Disadvantages:

- ➡ More complex to generate and manage.
- ➡ Requires specialized meshing tools to ensure smooth transitions.

Use in HEC-RAS:

- ➡ Often used when integrating 1D river models with 2D floodplain models.
- ➡ Helps improve accuracy in floodplain regions while keeping computational costs manageable.

1.5.4 Conclusion

In HEC-RAS, unstructured grids are most commonly used for 2D modeling due to their flexibility in handling complex terrains. However, structured grids can be beneficial for efficiency in regular channel sections. Hybrid grids are valuable when combining river and floodplain models to optimize accuracy and performance.

1.6 Experimental Validation and Calibration in CFD

Computational Fluid Dynamics (CFD) simulations provide a numerical approach to solving fluid flow problems, but their reliability depends on proper validation and calibration. Experimental validation ensures that the numerical model accurately represents real-world physics, while calibration fine-tunes the model parameters for improved accuracy. Without experimental validation, CFD predictions may be uncertain and lack credibility in engineering applications.

1.6.1 Importance of Experimental Data in CFD

Experimental data plays a crucial role in CFD simulations by providing a benchmark against which numerical results can be compared. This validation process helps ensure that the numerical methods, boundary conditions, and turbulence models accurately capture the physics of fluid flow.

1. Need for validation using experimental results.
2. Sources of experimental data (lab-scale and field-scale measurements).

1.6.2 Need for Validation Using Experimental Results

Validation is essential for:

- ➡ **Assessing Accuracy:**
  Comparing simulation results with experimental data helps quantify numerical errors and deviations.
- ➡ **Ensuring Model Reliability:**
  Establishing confidence in the CFD model before it is used for design, optimization, or decision-making.
- ➡ **Identifying Limitations:**
  Highlighting areas where improvements in grid resolution, turbulence modeling, or boundary conditions are required.
- ➡ **Regulatory Compliance:**
  Many industries, such as aerospace, automotive, and environmental engineering, require CFD results to be validated against experimental findings before approval.

In cases where significant discrepancies exist between CFD and experiments, model calibration is performed by adjusting parameters such as turbulence closure coefficients, wall roughness, or boundary conditions.

1.6.3 Sources of Experimental Data

Experimental data for CFD validation can be obtained from controlled laboratory-scale experiments or large-scale field measurements. Each source has its advantages and challenges.

- ➡ **A. Lab-Scale Measurements**
  Laboratory experiments are conducted under controlled conditions, allowing for high-precision data collection. Common setups include:

1. **Wind Tunnels:**
   Used in aerospace and automotive applications to measure aerodynamics and turbulence characteristics.
2. **Water Flumes & Hydraulic Channels:**
   Used for analyzing flow behavior over structures, sediment transport, and hydrodynamics.
3. **Particle Image Velocimetry (PIV):**
   A non-intrusive technique that provides detailed velocity field measurements in a flow.
4. **Laser Doppler Anemometry (LDA):**
   Used for pointwise velocity measurements with high accuracy.

Advantages:

1. High accuracy and repeatability.
2. Controlled boundary conditions enable systematic studies.

Challenges:

1. Scaling effects may cause differences between lab conditions and real-world scenarios.
2. Limited capability for replicating large-scale or highly complex systems.

➡ **B. Field-Scale Measurements**
Field experiments provide real-world data under natural conditions, making them highly valuable for validation. Common sources include:

1. **Environmental Monitoring:**
   Air and water quality measurements, pollutant dispersion studies, and meteorological observations.
2. **Full-Scale Wind Tunnel or Open-Field Tests:**
   Used in the assessment of building aerodynamics, wind energy studies, and urban air quality modeling.
3. **Flow Measurements in Rivers and Coastal Areas:**
   Acoustic Doppler Current Profilers (ADCP) and remote sensing techniques help capture large-scale flow patterns.

Advantages:

1. Realistic representation of operational conditions.
2. Suitable for complex and large-scale applications where lab replication is not feasible.

Challenges:

1. Difficult to control environmental variables (temperature, pressure, humidity, etc.).
2. Expensive and time-consuming compared to lab-scale experiments.

Both lab-scale and field-scale measurements contribute significantly to improving CFD simulations.

1.7 Basics of Modeling Using HEC-RAS

1.7.1 Introduction to HEC-RAS

HEC-RAS (Hydrologic Engineering Center's River Analysis System) is a powerful hydraulic modeling software developed by the U.S. Army Corps of Engineers. It is widely used for simulating open channel flow, floodplain mapping, and hydraulic structure analysis. HEC-RAS enables engineers and hydrologists to analyze steady and unsteady flow conditions, perform sediment transport simulations, and evaluate water quality impacts.

1.7.2 Key Features of HEC-RAS

HEC-RAS includes a range of features that make it suitable for hydraulic modeling:

- **Steady and Unsteady Flow Analysis:**
  Supports one-dimensional (1D) and two-dimensional (2D) flow modeling to simulate various hydraulic conditions.
- **Sediment Transport Modeling:**
  Predicts sediment erosion, transport, and deposition over time, which is essential for river and reservoir management.
- **Water Quality Analysis:**
  Assesses pollutant dispersion and water quality parameters such as temperature, dissolved oxygen, and nutrient concentrations.
- **Bridge and Culvert Analysis:**
  Evaluates hydraulic impacts of bridges, culverts, and other structures on river flow, aiding in flood risk assessment.
- **Floodplain Mapping:**
  Generates flood inundation maps that help in flood risk management and urban planning.
- **Dam Breach Analysis:**
  Simulates dam failure scenarios to assess potential downstream impacts and mitigation strategies.
- **Geospatial Integration:**
  Works with GIS software for improved spatial analysis and visualization of hydraulic simulations.

1.7.3 Steps in HEC-RAS Modeling

HEC-RAS modeling involves several key steps to ensure accurate and reliable simulations:

Define Geometry

1. Input river cross-sections, channel dimensions, and hydraulic structures.
2. Define bank stations and overbank areas.
3. Incorporate bridges, culverts, weirs, and levees where applicable.

Set Boundary Conditions

1. Specify inflow and outflow conditions based on observed or estimated hydrological data.
2. Define lateral inflows, tributary interactions, and hydraulic controls.
3. Assign roughness coefficients (e.g., Manning's n-values) to represent channel and floodplain conditions.

Assign Flow and Hydraulic Parameters

1. Specify steady or unsteady flow conditions depending on the analysis objectives.
2. Set computational timestep and numerical solvers for simulation accuracy.
3. Include precipitation, evaporation, and infiltration factors if required.

Run Simulations

1. Execute steady or unsteady flow analysis.
2. Perform sensitivity analysis to assess the influence of model parameters on results.
3. Adjust model settings to improve numerical stability and convergence.

Analyze Results

1. Interpret water surface profiles, velocity distributions, and depth variations.
2. Evaluate flood inundation maps for different return periods.
3. Assess hydraulic performance of structures under different scenarios.
4. Validate model results by comparing simulated outputs with observed data.

1.7.4 Model Calibration and Validation

Calibration and validation are essential steps to enhance model reliability:

- ➡ **Calibration:**
  Adjust parameters (e.g., roughness coefficients, boundary conditions) to match observed data.
- ➡ **Validation:**
  Compare simulation results with independent datasets to verify model accuracy.
- ➡ **Statistical Evaluation:**
  Use Nash-Sutcliffe efficiency, root mean square error (RMSE), and other metrics to assess model performance.

1.7.5 Applications of HEC-RAS

HEC-RAS is extensively used in various civil and environmental engineering applications, including:

- ➡ **Flood Risk Assessment:**
  Identifying flood-prone areas and designing mitigation measures.
- ➡ **River Restoration:**
  Evaluating the impacts of channel modifications and habitat restoration projects.
- ➡ **Infrastructure Design:**
  Analyzing the hydraulic performance of bridges, culverts, and embankments.

- **Emergency Management:**
  Assessing dam breach scenarios and developing evacuation plans.
- **Urban Drainage Modeling:**
  Simulating stormwater flow and urban flood events.

1.7.6 Future Trends in HEC-RAS Modeling

With advancements in computing and data science, HEC-RAS is evolving to incorporate:

- **Artificial Intelligence and Machine Learning:**
  Enhancing predictive accuracy and model automation.
- **Cloud-Based Computing:**
  Enabling large-scale simulations and real-time data integration.
- **Remote Sensing and GIS Integration:**
  Improving spatial analysis with high-resolution terrain and hydrological data.
- **3D Hydraulic Modeling:**
  Advancing towards three-dimensional (3D) simulations for more detailed flow analysis.

HEC-RAS continues to be a vital tool for hydraulic engineers and researchers, supporting informed decision-making in water resources management and flood risk assessment.

1.8 Uniform and Non-Uniform Flow in Open Channel Simulations

1.8.1 Uniform Flow:

- ➡ In uniform flow, the depth and velocity of water remain constant along the length of the channel.
- ➡ It typically occurs in long, prismatic channels where the slope, roughness, and cross-section remain unchanged.
- ➡ The driving force (gravity) and resisting force (friction) balance each other, resulting in steady flow conditions.
- ➡ A key example is normal flow in a straight river reach, which can be estimated using Manning's equation.

1.8.2 Non-Uniform Flow:

- ➡ Non-uniform flow occurs when water depth and velocity change along the channel.
- ➡ It can be classified into gradually varied flow (GVF) and rapidly varied flow (RVF).
- ➡ GVF occurs when there are gradual changes in flow depth due to backwater effects, obstructions, or mild channel slopes.
- ➡ RVF happens in cases of sudden changes, such as at hydraulic jumps, weirs, or spillways.

1.8.3 Using HEC-RAS for Uniform and Non-Uniform Flow Simulation

HEC-RAS is widely used for open-channel flow simulations, including both uniform and non-uniform flow scenarios.

Uniform Flow in HEC-RAS:

- ➡ Can be approximated by ensuring a long, prismatic channel with a steady flow regime.
- ➡ Manning's equation is used for normal depth calculations.
- ➡ The steady-flow analysis module in HEC-RAS can compute flow profiles under uniform conditions.

Non-Uniform Flow in HEC-RAS:

- ➡ The software's steady and unsteady flow analysis tools help model gradually and rapidly varied flows.
- ➡ Backwater curves, hydraulic jumps, and flow transitions can be simulated using energy and momentum equations.
- ➡ The unsteady flow module allows dynamic analysis of flood waves, dam breaks, and tidal effects.

1.8.4 Example: Simulating Uniform and Non-Uniform Flow in HEC-RAS

Scenario: A river reach with a mild slope and a downstream dam.

Uniform Flow Simulation:

- ➡ Define a long, straight river reach with a constant bed slope and cross-section.

- ➡ Set steady flow boundary conditions with constant discharge at the upstream end.
- ➡ HEC-RAS will compute a uniform flow profile using Manning's equation.

Non-Uniform Flow Simulation:

- ➡ Introduce a downstream dam, causing backwater effects.
- ➡ Specify varied flow conditions, such as increasing discharge or changing bed slopes.
- ➡ Run steady-flow analysis to compute gradually varied profiles or unsteady analysis for flood wave propagation.